Haoyu Zhai

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EDUCATION

- **University of Illinois at Urbana-Champaign** *Aug.2026 - Present*
  Ph.D. in Computer Science
  Advisor: Gang Wang
- **University of Illinois at Urbana-Champaign** *Aug.2024 - May 2026*
  M.S. in Computer Science
- **University of Illinois at Urbana-Champaign** *Aug.2020 - May.2024*
  B.S. in Mathematics and Computer Science, GPA: **3.96**/4.00
  B.S. in Statistics

RESEARCH INTERESTS

LLM Agent, Trustworthy AI, Model Optimization, Human-Centered AI Security, Reinforcement Learning

PUBLICATIONS

(*The authors contribute equally to this paper (co-first authors))

- [**ICML 2026**] Y. Chen, **H. Zhai**, C. Wang, R. Yang, L. Zhang, G. Wang, H. Zhang. **CAPTCHA Solving for Native GUI Agents: Automated Reasoning-Action Data Generation and Self-Corrective Training**. Proceedings of *the 43rd International Conference on Machine Learning*, July 2026
- [**IEEE SP 2026**] **H. Zhai***, S. Wang*, Q. Hao, P. Naghavi, G. Wang. **Revelio: Blurred Images Can Still Disclose Your Identity**. Proceedings of *the 47th IEEE Symposium on Security and Privacy*, May 2026
- [**NeurIPS 2025**] J. Liu*, N. Diwan*, Z. Wang*, **H. Zhai**, X. Zhou, K. A. Nguyen, T. Yu, M. Wahed, Y. Deng, H. Benkraouda, Y. Wei, L. Zhang, I. Lourentzou, G. Wang. **PurpCode-R1: Reasoning for Safer Code Generation**. Proceedings of *the 39th Annual Conference on Neural Information Processing Systems*, Dec 2025
- [**SOUPS 2025**] Y. Wang, **H. Zhai**, C. Wang, Q. Hao, N. A. Cohen, R. Foulger, J. A. Handler, G. Wang. **Can You Walk Me Through It? Explainable SMS Phishing Detection using LLM-based Agents**. Proceedings of *the 21st Symposium on Usable Privacy and Security*, August 2025

RESEARCH EXPERIENCE

- **Amazon Trusted AI Challenge: Trusted Software Agents**, UIUC *Nov.2025 - Present*
  - Formulated a novel threat model for utility-preserving vulnerability injection in coding agents, where adversarial simulated users induce security vulnerabilities while maintaining utility performance to maximize attack stealth.
  - Developed a **multi-agent simulated-user framework** using Strands Agents, with a planning agent orchestrating specialized sub-agents for adversarial reasoning, tool execution, state tracking, and adaptive interactions.
  - Constructed a real-world vulnerability benchmark and evaluation pipeline to measure attack success and functional utility, quantifying the tradeoff between vulnerability injection effectiveness and utility preservation.
- **Understanding User Perception of Deepfake Video Conference Calls**, UIUC *Aug.2025 - Present*
  - Built an end-to-end **deepfake video conferencing system** by integrating DeepFaceLab for face swapping and Retrieval-based Voice Conversion (RVC) for voice cloning.
  - Designed and conducted two human-subject studies comparing participants' perception and detection of deepfake presenters under unprompted real-time conferencing versus explicit video-classification prompts.
  - Identified a substantial gap in deepfake detection between prompted and unprompted settings, revealing the strong influence of user awareness on perceived authenticity and trust.
- **ReCAP: CAPTCHA Solving GUI Agent**, UIUC *Aug.2025 - May. 2026*
  - Contributed to the reasoning-action GUI agent design, including dynamic CAPTCHA task construction, automated trajectory generation, and self-correction data from failed attempts.

- Led evaluation across CAPTCHA-solving and general GUI-agent benchmarks, demonstrating strong CAPTCHA performance gains while preserving competitive general GUI interaction capability.

- **Divergence-Controlled RL for Multi-Hop Graph Reasoning**, UIUC *Aug.2025 - Dec. 2025*
  - Conducted a systematic study of **reinforcement learning optimized GraphRAG agents**, revealing how KL-divergence penalties regulate exploration–exploitation dynamics and stabilize GRPO training.
  - Designed and executed controlled experiments on the 2WikiMultiHopQA benchmark to quantify the interaction between divergence control, model scaling in the Qwen series, and multi-step graph-reasoning performance.

- **PurpCode-R1: Reasoning for Safer Code Generation**, UIUC *Jan.2025 - Aug.2025*
  - Contribute to the Amazon Nova AI Challenge (blue team) to develop reliable **LLM-based coding assistants**. Design high-coverage adversarial prompts as internal red team, simulating real-world unsafe coding scenarios.
  - Lead the end-to-end data curation pipeline: aggregated jailbreak prompts and templates from 10+ public safety datasets, apply LLM-based filtering to identify high-quality prompts, and generate aligned targets.
  - Benchmark model robustness against advanced search-based jailbreak methods (e.g., AutoDAN, GCG-Transfer), demonstrating superior performance over Qwen-series models across all evaluated attacks.

- **Face Image Deblur**, UIUC *Feb.2023 - May.2025*
  - Lead research on reconstructing intentionally blurred face images posted on social platforms, assessing the potential privacy leakage risks associated with existing blurring techniques.
  - Develop a multi-step approach integrating a conditional **diffusion model** for preliminary face restoration and an identity retrieval model to enhance fidelity using similar images.
  - Implement models in Python (**PyTorch**) and conducted large-scale experiments on public facial datasets, achieving **95.9%** recognition accuracy and outperforming state-of-the-art restoration methods.

- **LLM Phishing Agent**, UIUC *Aug.2024 - May.2025*
  - Design and implemented a robust **multi-agent LLM system** to detect SMS phishing, incorporating external knowledge (e.g., domain intelligence, webpage screenshots) to enhance reasoning and explainability.
  - Develop a **user-centric LLM agent** capable of delivering clear, actionable security advice, with tailored explanations optimized for elderly users through chain-of-thought prompting.
  - Achieve **98.8%** accuracy on real-world SMS phishing datasets; conduct a user study with 175 participants, earning a top-tier usability rating (SUS score: **82.6**).

WORK EXPERIENCE

- **Computer Network Information Center, Chinese Academy of Science** *May.2023 - Aug.2023*
  Big Data Developer Intern
  - Develop Lynx, a customizable Cypher query execution framework suitable for any type of database, allowing developers to query any type of database with graph database query syntax, avoiding costly database migrations.
  - Apply Lynx to **MySQL** and **MongoDB** using **Scala** and implement necessary APIs to connect mySQL database to Lynx, enabling graph query searching on relational databases and NoSQL databases.
  - Test the query performance using LDBC benchmark and optimize the framework by manipulating database indexes and improving code logic, reducing single record query time **from seconds to tens of milliseconds**.

HONORS AND AWARDS

- **Amazon Nova AI Challenge**, 1st Place Winner ($250K Prize) *Jul.2025*
- **Illinois Statistics Datathon**, Best Data Visualization & Top 10 Model Accuracy *Dec.2023*
- **Highest Distinction**, B.S. in Mathematics & Computer Science, UIUC *May.2024*
- **Highest Distinction**, B.S. in Statistics, UIUC *May.2024*
- **Dean's List**, College of Liberal Arts & Sciences, UIUC *2020-2023*

SKILLS

- **Programming Languages:** Python, R, C++, Java, SQL, Scala, JavaScript, HTML, Assembly, Shell
- **Technologies/Frameworks:** Agent frameworks (Strands Agents, LangGraph), LLM Frameworks (vLLM, VerL), Python Libraries (PyTorch, Pandas, Matplotlib, OpenCV), AWS, NoSQL, Flask, Git, Docker, Linux